

CORRECTIVE ACTION REPORT

CAR No.:	CAR-2023-001
Product:	H-Acid
Date Issued:	2023-01-20
Date Closed:	2023-05-20
Prepared By:	Supply Quality Engineer
Status:	CLOSED

1. PROBLEM STATEMENT

Iron content (Iron_ppm) in H-Acid batches B0351–B0430 exceeded the specification upper limit of 10 ppm. The I-MR SPC control chart (Minitab) flagged the first out-of-control signal at Batch B0355. Peak values reached 16.21 ppm. Process capability index Ppk for Iron_ppm dropped from 1.18 (Batches 1–350, pre-drift) to 0.29 (all 500 batches, including drift period), indicating the process was incapable of meeting specification during this interval.

2. CONTAINMENT ACTION

Immediate containment actions taken within 72 hours of SPC alarm:

- 100% iron ppm re-test of all Batches B0351–B0430 (QA lab ICP analysis).
- Quarantine and hold of suspect inventory pending re-test results.
- Customer hold notification issued for any shipments from affected period.
- Increased sampling frequency to every-batch testing during investigation.

3. ROOT CAUSE

Root Cause (Machine): Progressive corrosion and wall-thickness reduction of the cast-iron lining in Reduction Reactor R-02. Inspection confirmed >40% wall-thickness loss at the bottom wear zone. Under the acidic conditions of the iron/acid reduction step, corroded lining material leached directly into the batch.

Contributing Cause (Material): Supplier D (Iron Scrap) lot purity below internal spec (91.0% actual vs. 93.0% required), confirmed by incoming QA records. Lower-purity scrap accelerated the lining corrosion mechanism.

Fishbone Analysis: Confirmed via structured Cause-and-Effect diagram — 'High Iron Content in H-Acid Batches' — with primary causes in Machine and Material branches.

4. CORRECTIVE ACTION

CA-1: Replaced Reactor R-02 cast-iron lining with corrosion-resistant SS316L stainless steel (completed Feb 12, 2023 — MWO-2023-0212).

CA-2: Issued corrective action to Supplier D requiring ICP-verified CoA per lot, minimum incoming purity $\geq 94\%$, and quarterly re-audit. Supplier D Q4-2023 re-audit: purity 95.2%, OTD 89%, Score 83/100.

5. PREVENTIVE ACTION

- Reduced reactor lining inspection interval from 6 months to 3 months (bi-monthly PM).
- Updated Control Plan: every-batch Iron_ppm QA lab testing (vs. prior every-5th-batch sampling).
- Supplier Procedure SQP-004 updated: iron scrap suppliers must provide ICP CoA per incoming lot.

- Minitab I-MR chart updated with 8.5 ppm early-warning alert threshold (below USL=10 ppm).

6. VERIFICATION & EFFECTIVENESS

Post-correction monitoring — Batches B0431–B0500 (70 batches, Feb–May 2023):

- Iron_ppm Mean: 7.34 ppm | StdDev: 1.41 ppm | Max: 11.2 ppm
- Ppk (post-correction, 70 batches): 1.16 [Before corrective action (Batches 1-350): Ppk = 1.18]
- Zero additional OOC signals in 70 post-correction batches.
- Result: Process capability fully restored to pre-drift levels. Corrective action verified EFFECTIVE.
- CAR formally closed 2023-05-20 by Plant Quality Manager sign-off.

Synthetic data disclaimer: All numerical data is simulated for portfolio demonstration. No proprietary employer information is contained herein.